

Software Requirements Specification

AI-Driven Smart Recruitment & Interview Management System

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Version 1.0

Revision History

Name	Date	Reason For Changes	Version
SRIM	April 8, 2026	Initial Draft	1.0

1. Introduction

1.1 Review of Related Literature

The recruitment industry has evolved with platforms like HireVue and Modern Hire, which use AI and structured interviews to improve hiring decisions. Studies show that structured assessments are more effective than traditional interviews.

However, current systems still have limitations, such as static skill evaluation and limited feedback analysis. This project aims to improve the hiring process by providing a more structured and efficient recruitment system.

1.2 Problem Statement

HR departments at mid-to-large organizations face compounding inefficiencies during technical hiring: manual resume screening is time-intensive and subjective; interview scheduling across multiple evaluators creates coordination overhead; skill-gap analysis is inconsistent between interviewers; and post-interview feedback is fragmented across emails and spreadsheets. These inefficiencies lead to longer time-to-hire, higher cost-per-hire, and increased risk of overlooking high-potential candidates who do not fit traditional keyword filters.

1.3 Proposed Solution

The AI-Driven Smart Recruitment & Interview Management System (SRIM) is a multi-module web platform that automates the technical hiring lifecycle end-to-end. The solution introduces three interconnected modules: the HR Admin Module for requisition and pipeline management, the Technical Interviewer Module for structured evaluation, and the Candidate Module for application and assessment workflows.

1.4 Problem Scope

This SRS covers the SRIM platform in its initial release scope:

- Web application accessible via modern browsers (Chrome 110+, Firefox 110+, Edge 110+).
- Three user roles: HR Admin, Technical Interviewer, Candidate.
- Support for technical hiring workflows in software engineering.
- Integration with third-party calendaring (Google Calendar API) and email (SMTP).
- Out of scope: mobile-native apps, payroll integration, offer letter generation, and advanced video analytics beyond basic proctoring.

1.5 Challenges

- Designing a skill-weighting algorithm that generalizes across heterogeneous job roles without manual calibration per requisition.
- Ensuring proctoring accuracy while balancing candidate privacy and minimizing false-positive flags.
- Achieving real-time, low-latency feedback aggregation across distributed interviewer sessions.
- Building a bias-mitigation layer into AI-driven scoring to satisfy EEO compliance requirements.

1.6 Knowledge Areas Required

- Machine Learning & NLP: Resume parsing, skill extraction, behavioral signal analysis.
- Web Development: frontend, backend, REST API design.
- Database Engineering: MySQL schema design for recruitment workflows, Redis for caching.
- Security: Role-based access control (RBAC), data encryption, proctoring session management.
- Human-Computer Interaction: UX design for multi-persona interfaces.

1.7 Completeness Criteria

The project is considered complete when:

1. All three modules (HR Admin, Technical Interviewer, Candidate) are fully functional and integrated.
2. The Dynamic Skill-Weighting Engine produces ranked candidate lists with measurable correlation to interviewer ratings.
3. Proctored assessments operate with fewer than 5% false-positive integrity flags in user acceptance testing.
4. End-to-end system performance meets targets defined in Section 5.1.
5. All functional use cases described in Section 4 pass acceptance tests.

1.8 Research Outcomes / Nature of End Product

The end product is a deployable, full-stack web application demonstrating the efficacy of AI-assisted structured interviewing. Research contributions include: (a) a configurable skill-weighting model adaptable to dynamic job role definitions; (b) a proctoring heuristic combining computer vision and interaction telemetry; and (c) a feedback aggregation schema enabling structured comparative analysis across hiring cohorts.

1.9 Learning Outcomes

- Applying agile software development practices to a multi-module system.
- Implementing machine learning pipelines within a production web environment.
- Designing multi-tenant, role-segregated database schemas.
- Integrating third-party APIs (calendar, email) securely.
- Conducting user acceptance testing with domain stakeholders.

1.10 Document Conventions

Throughout this document:

- Bold text denotes module names, key terms, or requirement identifiers.
- Angle brackets < > indicate placeholder values to be replaced.
- UC-N denotes use case identifiers (e.g., UC-1, UC-2).
- FR-N denotes functional requirement identifiers.
- NFR-N denotes non-functional requirement identifiers.

2. Background Study and Literature Survey

Technical recruitment has evolved from paper-based resume screening to AI-assisted multi-stage pipelines. The following subsections review the relevant literature and existing systems.

2.1 Existing Platforms

HireVue (2004–present) pioneered on-demand video interviews with automated scoring. Its AI analyzes speech patterns and facial micro-expressions, though its predictive validity has been contested in peer-reviewed literature (Hickman et al., 2021). Modern Hire differentiates through structured competency frameworks aligned to O*NET occupational data. Both platforms, however, lack deeply configurable skill weighting tied to dynamic job requisitions.

2.2 AI in Recruitment

Raghavan et al. (2020) documented bias amplification in resume-screening algorithms trained on historical hiring data. Mitigation strategies include adversarial debiasing, fairness constraints in ranking (Zehlike et al., 2017), and demographic parity audits. This project incorporates a configurable fairness layer in the screening triage component.

2.3 Online Assessment Integrity

Proctoring research identifies three effective signal categories: visual (eye gaze, head pose), behavioral (typing cadence, copy-paste events), and environmental (background noise, browser activity). Combining these signals yields higher detection accuracy than any single modality (Alessio et al., 2017). SRIM adopts a lightweight composite heuristic suitable for browser-based deployment without requiring proprietary lockdown software.

2.4 Structured Interviewing

Meta-analyses consistently show that structured interviews (standardized questions, anchored rating scales, panel evaluation) outperform unstructured interviews in predicting job performance (McDaniel et al., 1994; Huffcutt & Arthur, 1994). SRIM operationalizes structure through its rubric builder and enforced scoring flows in the Technical Interviewer Module.

2.5 Limitations of Existing Work

Existing platforms share three common limitations that SRIM addresses: (1) skill weighting is static and must be manually reconfigured per requisition; (2) feedback aggregation does not support multi-dimensional rubrics with per-dimension weights; and (3) proctoring and live evaluation exist in separate products, requiring data export/import and creating integrity chain-of-custody gaps.

3. Overall Description

3.1 Proposed Solution

The system is designed as a three-tier web application consisting of a frontend, a backend API , and database. Redis is used for caching handles files like resumes.

Candidates apply and take assessments through the system. Applications are processed and scored, then shortlisted candidates are scheduled for interviews. After interviews, feedback is collected and analyzed to help HR make final hiring decisions.

3.2 User Classes and Characteristics

Three primary user classes interact with SRIM:

- **HR Admin:** Manages job requisitions, recruitment pipelines, interviewer assignments, and global platform settings. Technically proficient in HR processes; moderate computer literacy. Primary consumer of analytics dashboards and final hiring recommendations.
- **Technical Interviewer:** Conducts live technical evaluations using provided rubrics and coding environments. High technical expertise in the relevant domain. Interacts primarily with the evaluation interface during scheduled sessions.
- **Candidate:** Submits applications, completes screening assessments, and tracks application status. General computer literacy expected. Interacts primarily with the candidate portal through a browser.

3.3 Operating Environment

- **Client:** Modern web browsers (Chrome 110+, Firefox 110+, Edge 110+). No plugin installation required.
- **Database:** MySQL 8+ , Redis 7+.
- **Network:** Minimum 5 Mbps upload/download for video proctoring sessions. HTTPS enforced throughout.
- **Third-Party APIs:** Google Calendar API (scheduling), SMTP server (email notifications), OpenAI API or equivalent (NLP-based resume parsing).

3.4 Design and Implementation Constraints

- RBAC must enforce strict data isolation: candidates cannot access other candidates' data; interviewers can only view candidates assigned to their requisitions.
- The proctoring module must operate within browser sandbox constraints (WebRTC, Canvas API) without requiring system-level access.
- All personally identifiable information (PII) must be encrypted at rest (AES-256) and in transit (TLS 1.3+).
- The platform must comply with GDPR data retention and right-to-erasure provisions.
- AI scoring models must be auditable: all scoring inputs and weights must be logged and retrievable.

3.5 Assumptions and Dependencies

- **Assumption:** Candidate browsers support WebRTC and the MediaDevices API for webcam-based proctoring.

- Assumption: HR Admins will configure skill weights and rubrics before publishing job requisitions.
 - Dependency: Google Calendar API availability for scheduling integration.
 - Dependency: Availability of a suitable NLP model API (e.g., OpenAI GPT-4o or equivalent) for resume parsing and skill extraction.
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4. Functional Requirements

1. Automated Screening Triage:

Use Case Name	UC-1
Purpose	To automatically transition candidates through the hiring pipeline
Priority	High
Actors	Recruitment System , Candidate, HR Manager
Pre-conditions	Candidate has submitted an application; Job Requisition is active
Post-conditions	Candidate status is updated
Main Success Scenario	System receives new application
Alternative Scenario	System transitions state to "Rejected"

2. Dynamic Skill-Weighting Engine:

Use Case Name	UC-2
Purpose	calculate and assign a Score to candidates based on the weighted importance of specific technical and soft skills for a given job role
Priority	High
Actors	HR Manager , System (Calculator)
Pre-conditions	A Job Requisition exists. The candidate has submitted a resume or profile that the system has parsed
Post-conditions	The candidate is assigned a numerical Match Score, and their ranking in the applicant pool is updated
Main Success Scenario	1. HR defines required skills and sets weightings 2. System parses the incoming candidate profile 3. Engine aggregates the weighted scores into a final Match Score 4. System attaches the score to the candidate's profile for recruiters to view

Alternative Scenario	If the candidate's resume lacks parsable text , the System flags the profile with a "Pending Calculation" status and alerts HR to request a text-based format or trigger manual review.
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3. Job Requisition Approval Workflow:

Use Case Name	UC-3
Purpose	To ensure all new job openings clear a multi-tier logic gate of managerial and financial approvals before going live
Priority	Medium
Actors	HR Manager
Pre-conditions	HR Manager has drafted a complete Job Requisition including salary bands and required skills
Post-conditions	The job is either published live to job boards or returned to the draft state
Main Success Scenario	1. Hiring Manager submits the drafted requisition 2. System automatically publishes the job to external boards and internal portals
Alternative Scenario	An approver rejects the request due to budget constraints

4. Application Deduplication Logic:

Use Case Name	UC-4
Purpose	To identify and merge duplicate candidate applications across different roles or over time
Priority	High
Actors	System
Pre-conditions	A new application is submitted. The database contains existing candidate records
Post-conditions	The new application is either created as a brand-new profile or merged as a new application instance under an existing master profile
Main Success Scenario	1. System receives a new application 2. System appends the new job application and updated resume as a thread under the existing Master Candidate Profile

Alternative Scenario	It creates a new profile but flags it prominently as a prompting HR to manually merge or separate them
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5. AI-Ranked Shortlisting (Simulated):

Use Case Name	UC-5
Purpose	To automatically filter and rank the applicant pool based on keyword density, experience, and test scores, surfacing the top 10% to accelerate recruiter workflows
Priority	High
Actors	System
Pre-conditions	A batch of candidates exists for an active requisition. Resumes have been parsed, and initial Match Scores (UC-2) have been calculated
Post-conditions	Top candidates are highlighted on the recruiter's daily dashboard
Main Success Scenario	1. Algorithm calculates a combined score using keyword density, years of experience, and the UC-2 Match Score. 2. System places the top 10% into a "Shortlisted" tier
Alternative Scenario	If the total pool is too small to meaningfully calculate a top 10% , the System ranks all available candidates, skips the cutoff, and flags the requisition to the Recruiter to indicate that more sourcing efforts are needed.

6. External Job-Board Sync Manager:

Use Case Name	UC-6
Purpose	To automate the distribution of job postings to external job boards and track the origin of incoming applications
Priority	Medium
Actors	System , External Job Board APIs , HR Manager
Pre-conditions	Job Requisition is fully approved (UC-3) and its status is set to Published
Post-conditions	The job is live on selected external platforms, and any incoming applications contain a valid source tracking tag
Main Success Scenario	1. System detects a new "Published" job requisition

	2. When a candidate applies via a specific link, the System parses the parameters and logs the referral source (e.g., "LinkedIn") in the candidate's master profile
Alternative Scenario	The System retries , then flags that specific board as "Sync Failed" on the HR dashboard while keeping the successful channel connections active

7. Proctored Environment Controller:

Use Case Name	UC-7
Purpose	To monitor candidate behavior during a technical assessment to ensure test integrity and flag potential policy violations (like unauthorized assistance)
Priority	High
Actors	Candidate , System
Pre-conditions	Candidate has securely logged in and started an active assessment session within the platform's testing portal
Post-conditions	Assessment is completed; a timestamped "Integrity Log" is permanently attached to the candidate's test results
Main Success Scenario	1. System begins monitoring browser state (using the Page Visibility API). 2. Candidate switches tabs or clicks outside the assessment window. 3. System detects "Focus-Loss" and logs a timestamped event

8. Randomized Question-Bank Generator:

Use Case Name	UC-8
Purpose	To dynamically compile unique, fair, and balanced technical assessments for each candidate
Priority	High
Actors	System
Pre-conditions	A Job Requisition exists with defined required skills .The centralized Question Bank is sufficiently populated with tagged and categorized questions
Post-conditions	A unique, multi-part test instance is created and linked to the candidate's profile
Main Success Scenario	1. System receives a trigger to generate an assessment for a specific candidate and role 2. The assessment link is generated and dispatched to the candidate
Alternative Scenario	The System attempts to pull questions for a specific tag and difficulty, but the Question Bank lacks enough unused items to satisfy the criteria without repeating questions recently used for the same job requisition. The System pauses generation and alerts the Technical Hiring Manager to add more questions to the bank

9. Code-Execution Output Validator (Simulated):

Use Case Name	UC-9
Purpose	To automatically compile, execute, and grade a candidate's code submission by running it against a suite of hidden test cases to ensure the solution is robust and handles edge cases
Priority	High
Actors	System , Candidate
Pre-conditions	Candidate has written and submitted code for a specific algorithmic challenge. The system has a defined set of inputs and expected outputs for that challenge
Post-conditions	The submission is assigned a technical score, and execution logs are saved to the candidate's profile
Main Success Scenario	1. Candidate clicks "Submit Code." 2. System compiles (if necessary) and executes the code. 3. System compares the actual output against the expected output

10.Plagiarism Detection Logic (Simulated):

Use Case Name	UC-10
Purpose	To maintain assessment integrity by automatically identifying submissions that closely match known, publicly available solutions or previously submitted answers from other candidates
Priority	Low
Actors	System , Technical Reviewer
Pre-conditions	Candidate submits a technical assessment. The system's "Common Answer" database is populated with standard solutions
Post-conditions	The submission receives a "Similarity Confidence Score." If flagged, it is routed for manual review
Main Success Scenario	1. System receives the final code submission 2. System runs a similarity algorithm against the "Common Answer" database and the internal candidate history

11.Dynamic Difficulty Adjustment:

Use Case Name	UC-11
Purpose	To adaptively scale the difficulty of follow-up technical assessments based on a candidate's prior performance
Priority	Medium
Actors	System
Pre-conditions	Candidate has completed an initial baseline assessment (UC-8) and received a score. A multi-stage testing workflow is active
Post-conditions	A follow-up test is generated with a modified distribution of question difficulties
Main Success Scenario	1. System reviews the candidate's baseline test score 2. System triggers the generation of a "Stage 2" assessment
Alternative Scenario	If the candidate scores below the minimum threshold (<50%) on the baseline, the System does not generate a follow-up test. Instead, it triggers the "Rejected" workflow and ends the assessment phase

12.Assessment "Cool-down" Manager:

Use Case Name	UC-12
Purpose	To enforce a mandatory waiting period before a candidate can retake a specific technical assessment, preventing brute-force attempts and maintaining test integrity
Priority	High
Actors	System , Candidate
Pre-conditions	Candidate applies for a role that requires an assessment they have previously taken
Post-conditions	Candidate is either granted access to the test or blocked with an explanatory notification
Main Success Scenario	1. Candidate reaches the assessment stage of a new application 2. System checks the candidate's master profile for prior assessment timestamps of the exact same test type 3. System calculates the time elapsed since the last attempt 4. The elapsed time exceeds the defined cool-down period 5. System generates a new assessment link and allows the candidate to proceed
Alternative Scenario	The candidate took the test 2 months ago. The System blocks the generation of a new test, automatically pulls the historical score to use for the current application, and notifies the candidate that their previous score will be utilized due to the active cool-down period

13.Interviewer Availability Conflict Resolver:

Use Case Name	UC-13
Purpose	To automatically cross-reference multiple calendars and time zones to find mutually agreeable interview slots without manual HR coordination
Priority	High
Actors	System , Candidate , Technical Interviewer
Pre-conditions	Candidate is moved to the "Interview" stage. Interviewers' work calendars (Google Workspace) are integrated with the system
Post-conditions	A calendar invite with a video link is created and dispatched to all parties
Main Success Scenario	1. System prompts the Candidate to select 3 preferred time blocks 2. System normalizes the Candidate's selected times into UTC 3. System automatically generates a calendar invite with conferencing details and books the slot
Alternative Scenario	None of the Candidate's chosen times align with the Interviewers' schedules. The System automatically generates a new list of times where the <i>Interviewers</i> are free, converts them to the Candidate's local time zone, and emails the Candidate a link to select one of these guaranteed slots

14.Multi-Representative Panel Builder:

Use Case Name	UC-14
Purpose	construct interview panels requiring a mix of technical and /HR representatives
Priority	Medium
Actors	System
Pre-conditions	An interview stage requiring a panel is triggered. Employee profiles are tagged with their department and seniority level
Post-conditions	A balanced panel of interviewers is locked in for the session
Main Success Scenario	1. System receives a request to build a panel for a Mid-Level Backend Engineer role 2. System queries the employee database for one available "Senior Technical" , one available "HR representative" and groups these two individuals into a temporary Panel Object
Alternative Scenario	The System cannot find an available "Senior Technical" representative within the required timeframe. The System flags the panel as "Incomplete" and alerts the HR Coordinator to manually override the requirement or expand the search timeframe

15. Automated Interview Briefing Generator:

Use Case Name	UC-15
Purpose	To prepare for the interview by automatically compiling candidate resumes, test scores, and job requirements into a single, organized "Interviewer Pack"
Priority	Medium
Actors	System
Pre-conditions	Candidate resumes and test scores must already be stored in the system.
Post-conditions	A complete "Interviewer Pack" is generated and accessible before the interview.
Main Success Scenario	1.System retrieves candidate resume, test scores, and job requirements. 2.System compiles the data into a structured "Interviewer Pack."
Alternative Scenario	Candidate resume or test scores are incomplete System generates a partial pack and flags missing information.

16. Live Coding Environment Sync:

Use Case Name	UC-16
Purpose	To enable real-time collaboration during technical interviews by ensuring both interviewer and candidate see the same code changes instantly
Priority	Medium
Actors	System ,Candidate, Technical Interviewer
Pre-conditions	Coding environment must be initialized and connected.
Post-conditions	Both parties see code changes in real-time instantly
Main Success Scenario	1. Candidate begins coding in the environment 2. System mirrors candidate's screen and actions. 3. Interviewer observes candidate's coding, navigation, and test runs.
Alternative Scenario	1.Candidate disables shadowing 2,Interviewer receives notification.

17. Interviewer "Shadowing" Logic:

Use Case Name	UC-17
Purpose	To allow junior staff or trainees to observe interviews for learning purposes without their evaluations influencing the candidate's final score.
Priority	Low
Actors	System ,Candidate, Interviewer , Junior Staff
Pre-conditions	Interview must be scheduled in the system. Junior staff must have shadowing permissions enabled
Post-conditions	Junior staff successfully observe the interview. Candidate's official score reflects only the interviewer's assessment Observer notes or scores are recorded separately but excluded from the candidate's final evaluation.
Main Success Scenario	1.Junior staff join the interview session as observers. 2. System ensures observer scores are stored separately and not factored into the candidate's final average which is determined by only Interviewer's score
Alternative Scenario	Observer Attempts to Submit Score: System accepts but flags it as "training only" and excludes it from final score.

18. Session Extension Protocol:

Use Case Name	UC-18
Purpose	To provide a controlled mechanism for extending interview sessions when technical issues occur, ensuring fairness to candidates
Priority	High
Actors	System , Interviewer
Pre-conditions	Technical issues must be detected or reported and admin must be available to review extension requests.
Post-conditions	If approved, the session continues with added time and If denied, the session ends at the scheduled time.
Main Success Scenario	1. Technical issue occurs during the interview. 2.Interviewer requests a session extension through the system 3.Candidate and interviewer continue the interview seamlessly.
Alternative Scenario	Technical Issue Resolved Quickly: Interviewer cancels extension request before admin approval

19. Interviewer Load Balancer:

Use Case Name	UC-19
Purpose	To automatically distribute interviews among staff preventing burnout,
Priority	Medium
Actors	System
Pre-conditions	Interviewer workload data must be tracked (number of interviews, time commitments). Load balancing logic must be active and configured.
Post-conditions	Interviews are assigned to the interviewer with the lowest workload.
Main Success Scenario	1.HR schedules a new interview in the system. 2.System identifies the interviewer with the lowest current workload. 3.Interview is automatically assigned to that interviewer.
Alternative Scenario	Equal Workload: Multiple interviewers have the same workload System assigns the interview randomly

20. Multi-Dimensional Feedback Aggregator:

Use Case Name	UC-20
Purpose	To compile and unify scores from different interview dimensions into a single comprehensive report
Priority	High
Actors	System
Pre-conditions	Candidate interviews must be completed across different categories and Interviewers must submit their scores into the system.
Post-conditions	A unified candidate report is generated. Scores from different interviewers are compiled and normalized.
Main Success Scenario	1.Candidate completes all different types of interviews 2. Each interviewer submits their scores into the system 3. System generates a unified report with overall score
Alternative Scenario	Missing Scores: One interviewer fails to submit so, System generates partial report and flags missing data.

21. Score Normalization Algorithm:

Use Case Name	UC-21
Purpose	To ensure fairness in candidate evaluation by adjusting scores based on interviewer scoring trends
Priority	High
Actors	System
Pre-conditions	Interviewers must submit raw scores for candidates and historical scoring data must be available to detect interviewer trends.
Post-conditions	HR receives a fair and consistent report for decision-making.
Main Success Scenario	1.Interviewers submit candidate scores across different categories. 2.System analyzes historical scoring patterns for each interviewer 3.System detects deviations 4.System generates a final normalized report for HR.
Alternative Scenario	Insufficient Data: Not enough historical scores to detect trends System uses raw scores only.

22. Candidate "Red-Flag" Escalation:

Use Case Name	UC-22
Purpose	To immediately flag serious concerns about a candidate (e.g., ethical violations, inappropriate behavior, legal issues) for urgent HR review.
Priority	High
Actors	System , HR manager
Pre-conditions	An interview or assessment must be in progress and if Candidate behavior or data triggered a serious concern HR must be available to receive escalation alerts.
Post-conditions	A high-priority alert is sent to HR and Candidate's evaluation is paused until HR reviews.
Main Success Scenario	1.Interviewer or AI system detects a serious issue (e.g., candidate falsifies credentials). 2.System raises a "Red-Flag" alert. 3.HR receives immediate notification.
Alternative Scenario	1.System Error: Escalation fails 2. HR notified manually by interviewer.

23. Consensus Meeting Automator:

Use Case Name	UC-23
Purpose	To automatically trigger a debrief meeting only once all panel members have submitted their individual feedback.
Priority	Medium
Actors	System , Interviewers
Pre-conditions	Candidate interviews must be completed and all panel members must have access to submit feedback. Meeting scheduling logic must be configured in the system
Post-conditions	A debrief meeting is scheduled only after all feedback is submitted.
Main Success Scenario	1.Candidate completes interviews with multiple panel members. 2.Each panel member submits their feedback individually. 3.Once all feedback is received, system automatically triggers a debrief meeting.
Alternative Scenario	Incomplete Feedback: One or more panel members fail to submit so, System delays meeting and sends reminders.

24. Competency Gap Visualizer:

Use Case Name	UC-24
Purpose	To visually compare a candidate's actual interview scores against the "Ideal Profile" for a role using a spider-chart
Priority	High
Actors	System , HR Manager
Pre-conditions	Candidate must have completed all required interview rounds. Interviewers must submit scores for each Ideal profile benchmarks must be defined for the role.
Post-conditions	A spider-chart is generated showing candidate vs. ideal profile.
Main Success Scenario	1.Candidate completes interviews across multiple competencies. 2.Interviewers submit scores 3.System retrieves the "Ideal Profile" benchmarks 4.System generates a spider-chart comparing candidate scores vs. ideal profile.
Alternative Scenario	Missing Scores: One competency score not submitted Chart generated with partial data, flagged for HR.

25. Hiring Recommendation State-Machine:

Use Case Name	UC-25
Purpose	To formalize the decision-making process by transitioning a candidate through defined states(Strong Hire, Hire, No Hire, Strong No Hire)
Priority	High
Actors	System , HR Manager
Pre-conditions	Candidate must have completed all required interview rounds. Feedbacks and scores must be submitted
Post-conditions	Candidate is assigned a final recommendation state.
Main Success Scenario	1.Candidate completes interviews across multiple competencies. 2.Feedbacks and scores must be submitted by all interviews 3.System aggregates and normalizes scores. Based on thresholds and consensus rules: -If scores are excellent → Transition to Strong Hire . -If scores are good → Transition to Hire . -If scores are poor → Transition to No Hire If scores are very poor or red-flagged → Transition to Strong No Hire .
Alternative Scenario	Incomplete Feedback: Candidate remains in "Evaluated" until all feedback is submitted.

26. Offer Package Calculator:

Use Case Name	UC-26
Purpose	To automatically calculate a candidate's compensation package (salary, signing bonus, stock options) based on their level
Priority	High
Actors	System , HR Manager
Pre-conditions	Candidate must have a finalized hiring recommendation (e.g., Hire or Strong Hire).Role level and compensation rules must be defined in the system.
Post-conditions	A complete offer package is generated Candidate receives a formal offer letter
Main Success Scenario	1.Candidate is marked as “Hire” or “Strong Hire.” 2.System applies compensation rules 3.System generates a full offer package 4.HR reviews and sends the offer to the candidate.
Alternative Scenario	System Error: Calculator fails, HR notified, fallback to manual calculation.

27. Digital Offer-Letter Generator:

Use Case Name	UC-27
Purpose	To automatically generate a legally compliant offer letter by populating a standard template with candidate-specific data (name, role, compensation, start date), and prepare it for electronic signature
Priority	High
Actors	System , HR Manager
Pre-conditions	Compensation package must be calculated and legal offer letter template must be stored in the system.
Post-conditions	A complete digital offer letter is generated , the candidate receives the letter for electronic signature and signed copy is stored in the system for compliance.
Main Success Scenario	1.Candidate is marked as “Hire.” 2.System populates the legal offer letter template with candidate-specific data. 3.System generates a digital document ready for e-signature and candidate signs electronically.
Alternative Scenario	Missing Data: Candidate details incomplete System flags HR to fill missing fields before generation.

28. Offer Validity Timer:

Use Case Name	UC-28
Purpose	To enforce a defined “Decision Window” for candidates to accept or decline an offer. If the offer is not signed within that window, the system automatically expires it
Priority	Medium
Actors	System , HR Manager, Candidate
Pre-conditions	Candidate must have received a digital offer letter, a decision window must be defined and System must have access to timestamp data for when the offer was sent.
Post-conditions	If signed within the window , Offer accepted and logged. If not signed within the window , Offer automatically expires.
Main Success Scenario	1.Candidate receives a digital offer letter at 10:00 AM, April 9, 2026 . 2.System starts a 48-hour countdown. 3.Candidate signs the offer within the window. 4. System marks the offer as Accepted.
Alternative Scenario	No Response: Candidate does not sign → within the widow, system marks offer as Expired.

29. Counter-Offer Negotiation Tracker:

Use Case Name	UC-29
Purpose	To track approvals, and rejections during the salary negotiation phase in a structured and object-oriented way
Priority	High
Actors	System , HR Manager, Candidate
Pre-conditions	A candidate has already received an initial offer package
Post-conditions	The negotiation log is complete and accessible for compliance review
Main Success Scenario	-HR sends initial offer to candidate. -Candidate requests higher salary -HR reviews and either approves, rejects, or proposes another offer -Negotiation continues until final approval or termination.
Alternative Scenario	HR Rejects All Offers: Candidate marked as “Declined.”

30. Referral Reward Attribution:

Use Case Name	UC-30
Purpose	To automatically identify the employee who referred a hired candidate and trigger their referral bonus payout after the candidate successfully completes a defined probationary period
Priority	Low
Actors	System , HR Manager, Candidate , Interviewer
Pre-conditions	Candidate must be hired through the referral program. Referring employee must be recorded in the system at the time of candidate application Candidate must complete a defined probation period of employment
Post-conditions	System confirms probation completion and referral bonus is triggered for the referring employee.
Main Success Scenario	1.System records the referring employee's ID. 2.Candidate is hired and probation period begins. 3.After probation days, system verifies candidate is still employed. 4.Referral bonus is automatically triggered for the referring employee.
Alternative Scenario	Candidate Leaves Early: Candidate resigns or is terminated before probation days so, Bonus not triggered

31. Background Check Integration (Simulated):

Use Case Name	UC-31
Purpose	To automate the initiation, tracking, and resolution of external verification processes, ensuring legal and company compliance before finalizing a hire
Priority	High
Actors	System
Pre-conditions	Candidate has accepted a conditional job offer and signed a digital consent form for a background check
Post-conditions	Candidate is either cleared for onboarding or flagged for a compliance review
Main Success Scenario	1.System formats candidate data (Name, SSN, Address) into a secure payload 2.System transmits the payload via API to the external background check provider 3.External API sends a webhook confirming a "Pass" result 4.System automatically updates the status to "Cleared" and triggers the Pre-Onboarding workflow (UC-32)
Alternative Scenario	The external API returns a "Review Required" or "Fail" status .The System immediately halts automated onboarding, locks the candidate's

	progression, and alerts the HR/Compliance Officer to manually review the detailed report and make a final determination
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32. Template Versioning Manager:

Use Case Name	UC-32
Purpose	To manage the lifecycle of HR documents ensuring all staff access only the most current, approved versions while maintaining an audit trail of historical changes
Priority	Medium
Actors	System , HR
Pre-conditions	A central repository for templates exists within the system. Users have appropriate role-based access permissions
Post-conditions	A new version of a template is published globally; the previous version is deprecated and locked from future use
Main Success Scenario	1. HR Admin checks out the active 2. System automatically updates the status of v1.0 3. System updates all dynamic links within active Job Requisitions and upcoming Interviewer Briefing Packs (UC-15) to point to the new v2.0 template
Alternative Scenario	An interviewer is actively filling out the v1.0 rubric during a live interview at the exact moment the HR Admin publishes v2.0. The System detects the active session, locks the version for that specific instance to prevent data loss or scoring inconsistencies mid-interview, and later flags the submitted assessment with a note for HR review

5. Nonfunctional Requirements

5.1 Performance & Scalability

Pipeline Throughput Analytics:

-Logic that measures how efficiently candidates move through the recruitment funnel. It calculates metrics like *Time-to-Hire* and identifies whether there is too much delay in interview scheduling or offer approvals. It gives the HR the spot to notice the hiring delay and reduces it.

Database Integrity Manager:

-A background process that periodically archives old records (like "Closed" job requisitions and "Rejected" candidates) to keep the active database lean which prevents the system from slowing down giving a better performance

-As the data grows system remain scalable.

5.2 Reliability & Availability

Timed-Session Heartbeat:

-A background monitoring service that checks active test sessions and automatically submits them when the timer expires. Which guarantees that timed assessments always end correctly even if the browser crashes.

-Maintains consistent behavior across all users.

Automated Notification Escalator:

-An observer-pattern implementation that monitors pending feedback tasks. If an interviewer hasn't submitted feedback within a defined time window, the system automatically sends reminders or escalates the notification to higher levels (HR manager) which ensures critical workflow steps are not delayed, maintaining the reliability of the recruitment process.

-Therefore it ensures hiring decisions progress smoothly.

5.3 Security

Role-Based Access Control (RBAC):

-Ensures that interviewers can only see the candidates assigned to them while HR can see all candidates which prevents unauthorized access.

Data Retention & "Right to be Forgotten":

-function that define how long candidate data is stored and how it can be deleted which protects user privacy and reduces legal risk for the organization.

System Audit Trail:

-A permanent log of all system activities, such as status changes and score modifications ensuring that every action is traceable, captures (who, what changed and why)

5.4 Usability & Experience

Post-Interview Sentiment Logger:

- Collects feedback from candidates after interviews to measure their overall experience. The function is to track candidate sentiment so the company can spot strengths and weaknesses in its hiring process and build a stronger recruitment brand.
- This can improve usability.

5.5 Compliance & Legal

Data Retention & "Right to be Forgotten":

- Allows the system to delete candidate data after a set period to comply with privacy laws. Which ensures the protection of candidate privacy and meet legal obligations.
- Builds trust with candidates. Avoids fines and legal issues.

Diversity & Inclusion Audit Reporter:

- Generates reports on applicant diversity while keeping individual data anonymous.
- Supports fair recruitment and strengthens employer brand.
- This provides insights without exposing personal identities.
- This ensures there is no biases towards anything

8. Revised Project Plan

Task	Duration	Status	Deadline
Requirements & Analysis	Week 1–2	Completed	Week 2
System Architecture Design	Week 1-2	Completed	Week 2

9. References

[1] HireVue. (2024). AI-Powered Video Interviewing Platform. Retrieved from <https://www.hirevue.com/>

[2] Modern Hire. (2024). Structured Hiring Technology. Retrieved from <https://www.modernhire.com/>

10. Appendix: Glossary

- SRIM:** Smart Recruitment & Interview Management System — the platform described in this document.
- NLP:** Natural Language Processing — used for AI-driven resume parsing and skill extraction.
- RBAC:** Role-Based Access Control — a security model restricting system access based on user roles.
- Dynamic Skill-Weighting Engine:** The AI component that assigns configurable weights to required skills and computes weighted composite candidate scores.
- Proctoring:** The process of monitoring candidate behavior during online assessments to ensure integrity.
- GDPR:** General Data Protection Regulation — EU regulation governing personal data processing.
- EEO:** Equal Employment Opportunity — legal framework requiring non-discriminatory hiring practices.
- WebRTC:** Web Real-Time Communication — browser API enabling real-time audio, video, and data transmission.
- UC:** Use Case — a description of how a user interacts with the system to achieve a goal.